

Doping Stability Report -- Fe in KCoO2 at 300 C

Generated: 2026-05-30 13:37 Host: KCoO2 Dopant: Fe

Executive Summary

Fe preferentially occupies the Co layer. The formation energy difference between the two layers is +6.242 eV, strongly favouring the Co site. MD confirms this preference: on the Co site the dopant MSD plateau slope is 0.00356 Å²/ps (stable), compared to 0.00400 Å²/ps on the K site. The Co-site result is isovalent (no charge mismatch), so the formation energy is fully reliable. The K-site E_f is approximate due to a charge surplus of +2 that cannot be modelled in CHGNet.

Site	E_f (eV)	Charge	Compensation	MSD pass	Coord pass	Verdict
Co	-1.042	+0	No	PASS	PASS	STABLE
K	+5.200	+2	Yes	PASS	FAIL	STRUCTURAL COLLAPSE

Fe at the Co layer -- STABLE

Charge situation

Fe replaces Co with no change in oxidation state (isovalent substitution, charge mismatch = 0). No charge compensation is needed. The formation energy is fully reliable.

Formation Energy (thermodynamic stability)

Formation energy E_f	-1.0422 eV	PASS	< 1.0 eV to pass
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E_f = -1.042 eV. This value is strongly negative, indicating that Fe incorporation is thermodynamically very favorable at the Co site.

Molecular Dynamics Stability (300 C, 25 ps)

MSD slope (late-time)	0.00356 Å²/ps	PASS	< 0.005 to pass (plateau = stable)
Final MSD	0.223 Å²	----	
Max displacement	0.785 Å	----	
Coordination (min/mean/max)	4 / 5.0 / 5	PASS	relaxed = 5, tolerance +/-1
Mean bond length (MD vs relaxed)	1.988 Å (relaxed: 1.9)	----	
Lattice volume change	+0.00 %	PASS	< +/-5% to pass

Late-time MSD slope = 0.00356 Å²/ps. The slope is small and below the migration threshold -- the dopant is site-stable. Final MSD = 0.223 Å²; maximum displacement from starting position = 0.79 Å.

Coordination fluctuated between 4 and 5 (mean 5.0), within the +/-1 tolerance of the relaxed value (5).

This is consistent with normal thermal vibration. Mean nearest-neighbour distance during MD = 1.988 Å.

Volume change = +0.00%. The host lattice is mechanically intact.

VERDICT: STABLE

Fe is thermodynamically favorable and kinetically stable on the Co site at 300 C. All four stability criteria

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pass: formation energy is favorable, the dopant does not migrate during MD, its coordination environment is preserved, and the host lattice volume is unchanged.

Fe at the K layer -- STRUCTURAL COLLAPSE

Charge situation

Fe replaces K with a charge surplus of +2. CHGNet cannot model this compensation (would require Co oxidation or electron addition). The formation energy below is approximate.

Formation Energy (thermodynamic stability)

Formation energy E_f	+5.2002 eV	FAIL	< 1.0 eV to pass
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E_f = +5.200 eV (approximate -- charge surplus not modelled in CHGNet). This value is large and positive, indicating that Fe incorporation is thermodynamically unfavorable under equilibrium conditions at the K site.

Molecular Dynamics Stability (300 C, 25 ps)

MSD slope (late-time)	0.00400 Å²/ps	PASS	< 0.005 to pass (plateau = stable)
Final MSD	0.615 Å²	----	
Max displacement	1.236 Å	----	
Coordination (min/mean/max)	3 / 4.2 / 6	FAIL	relaxed = 5, tolerance +/-1
Mean bond length (MD vs relaxed)	1.928 Å (relaxed: 2.0)	----	
Lattice volume change	+0.00 %	PASS	< +/-5% to pass

Late-time MSD slope = 0.00400 Å²/ps. The slope is small and below the migration threshold -- the dopant is site-stable. Final MSD = 0.615 Å²; maximum displacement from starting position = 1.24 Å.

Coordination fluctuated between 3 and 6 (mean 4.2), deviating by more than +/-1 from the relaxed value (5). This may indicate structural distortion, or that the bond-length cutoff is too close to the actual bond length -- check the coordination_cutoff_A setting. Mean nearest-neighbour distance during MD = 1.928 Å.

Volume change = +0.00%. The host lattice is mechanically intact.

VERDICT: STRUCTURAL COLLAPSE

The coordination criterion failed while MSD and volume both pass, suggesting a possible cutoff artifact rather than a true structural collapse. The Fe atom barely moves (MSD = 0.615 Å²) but the coordination count fluctuates because the bond length at the K site is close to the analysis cutoff. Consider increasing coordination_cutoff_A and re-running analysis.

Site Preference Comparison

The Co site is preferred by 6.242 eV over the K site. Formation energy: -1.042 eV (Co) vs +5.200 eV (K).

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Caveats and Limitations

? CHGNet is charge-neutral.

The potential does not model oxidation states, polaron formation, or electron/hole localisation. Formation energies for aliovalent dopants (charge mismatch $\neq 0$) are approximate even when structural compensation defects are included.

? No Freysoldt/Kumagai correction.

Electrostatic finite-size corrections for charged defects require the host dielectric tensor and the DFT electrostatic potential, neither of which is available from a neural-network potential. For publication-quality E_f of charged defects, follow up with DFT using doped or pymatgen-analysis-defects.

? Metal-rich reference state.

Chemical potentials are computed from elemental ground-state structures. Synthesis under oxide-rich or other conditions shifts all E_f values by additive constants; the relative ordering of sites is unaffected.

? MD timescales are short (25 ps default).

This is sufficient to assess local site stability and rapid migration, but cannot resolve slow diffusion mechanisms. A non-plateauing MSD confirms instability; a plateau does not guarantee stability on longer timescales.

? Energy rankings are more trustworthy than absolute values.

CHGNet is a universal MLIP trained on r2SCAN DFT. Formation-energy errors for transition-metal oxides are typically 50-150 meV/atom. Use relative orderings (which site is preferred?) as the primary result.